



ALL	TASK NAME	DURATION	PLANNED START...	PLANNED FINISH ...	ASSIGNED	1, 20 '22	MAR, 27 '22	APR, 3 '22
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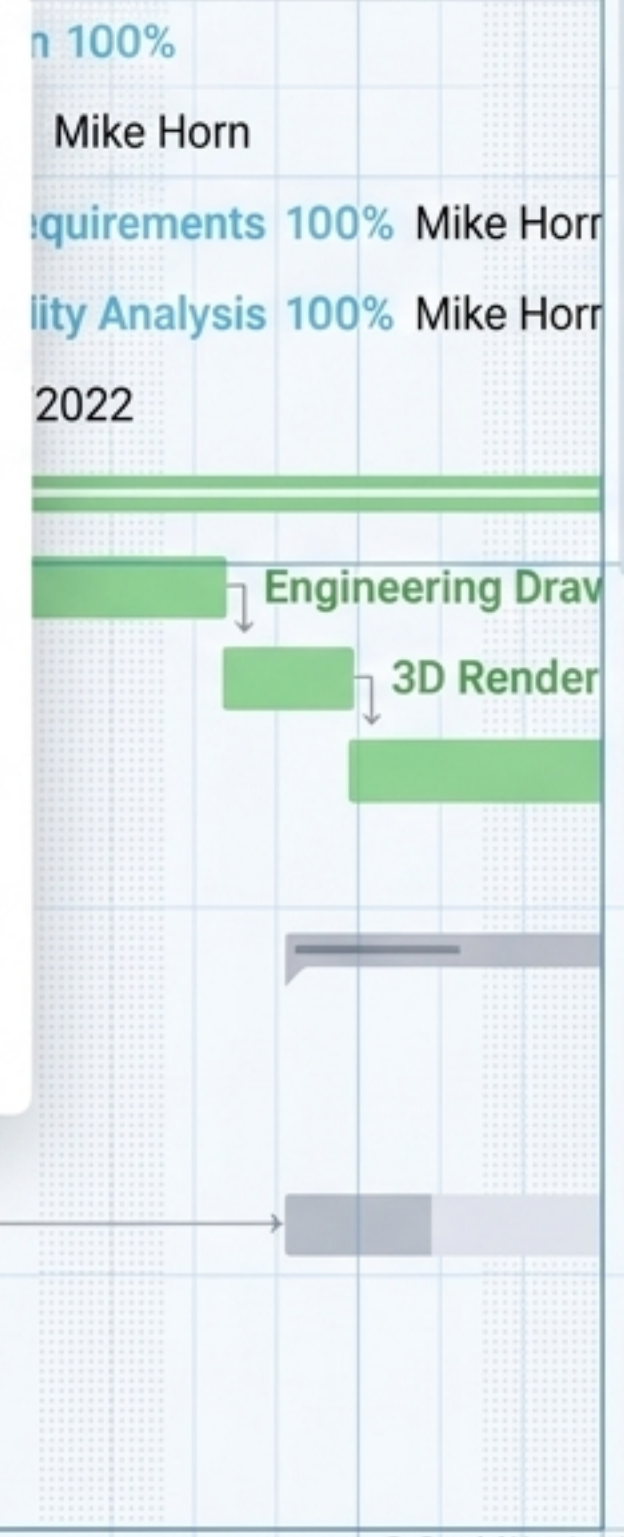


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2	Market Re
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4	Feasibilit
5	Stakehol
6	☐ Design
7	Engineeri
8	3D Rende
9	3D Printe
10	Stakehol
11	☐ Prototyping
12	CNC Part
13	Product Assembly
14	Durability & Stress T...
15	☐ Pre-production
16	New Production Eq...
17	Supply Chain Sourci...

SYS_MODULE_01-03

IT Project Management: Concept to Execution

Midterm Study Guide & Practice Workbook



Module 1: The Foundations

Definitions & Scope

Defining IT projects, comparing programs vs. portfolios, and the 10 PMBOK knowledge areas.



Module 2: Defining Success

Scope & SMART Objectives

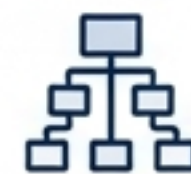
Expanding project goals into testable, specific, and measurable success criteria.



Module 3: Deconstructing the Work

WBS Mapping

Applying the 100% rule to build a 3-level Work Breakdown Structure for software delivery.



The “What”

Project: An endeavor to achieve an objective with a limited set of resources and a finite timeline.

IT Project: Specifically deals with IT infrastructure, information systems, or computers (e.g., software development, hardware installation, IT emergency recovery).

The "Why it's unique" (The IT PM Challenge)



- Must manage wide-scope interdependent integrations.

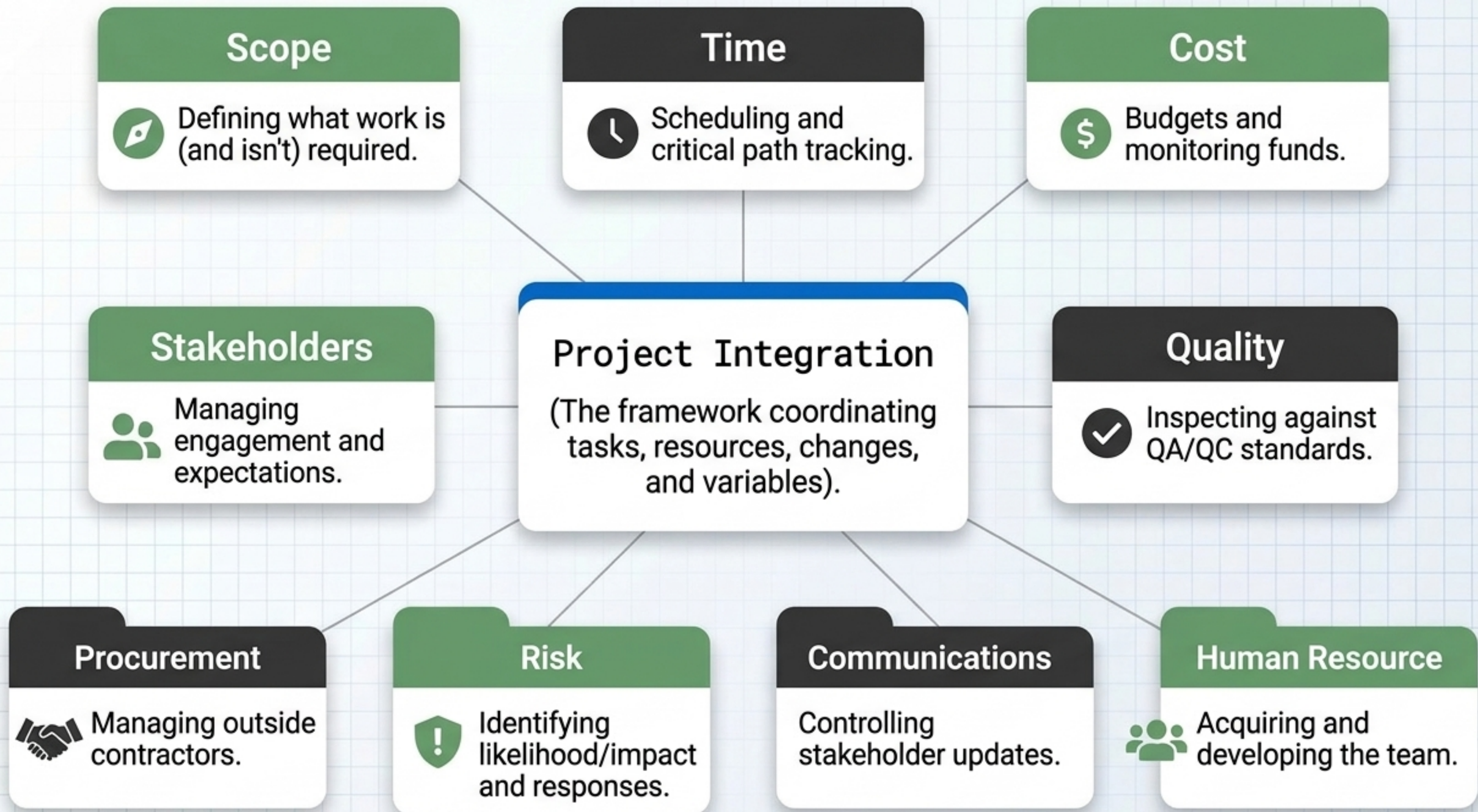


- Highly susceptible to software updates and scope creep.



- Requires specialized software (Kanban, online Gantt charts).

	Project Management	Program Management	Project Portfolio Management
Focus	Single endeavor.	Group of related projects.	Wider strategic scope.
Structure	Completing specific tasks on a finite timeline.	Executed simultaneously to optimize resource allocation.	Multiple projects and programs managed simultaneously (often under a PMO).
Objective	Deliver a specific output or objective.	Overcome shared unique challenges.	Achieve high-level business objectives and strategic goals.



1

Phase 1: Initiation (The 'Why')

Objective: Identify the business need.

Outputs: Project proposal, business plan, feasibility study.

2

Phase 2: Definition (The 'What')

Objective: Finalize requirements and outline the project plan.

Outputs: Finalized objectives, project scope, defined budgets, and resource allocation.

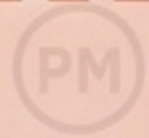
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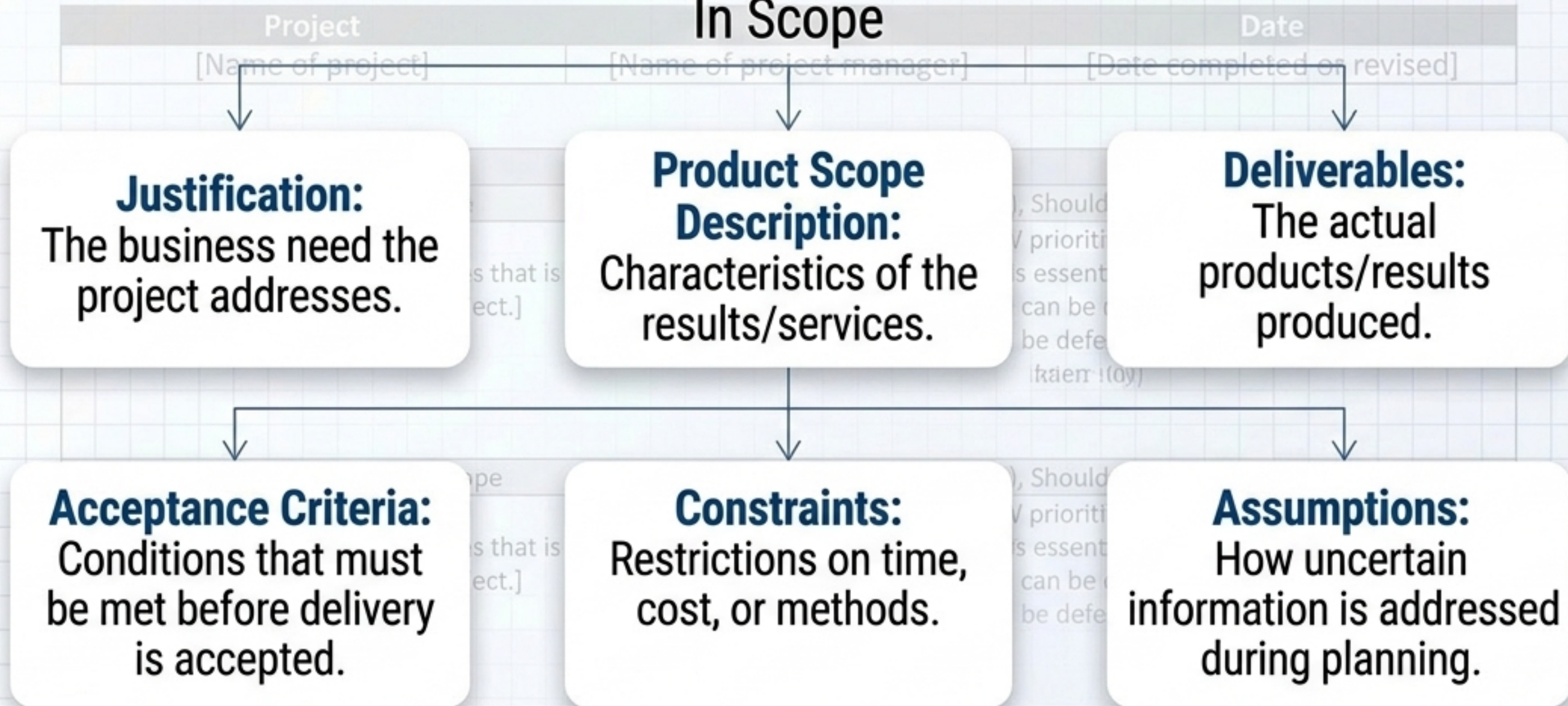
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Note: These upfront planning phases dictate the success of execution, monitoring/controlling, and closure.



Project Exclusions (Out of Scope)

In Scope



Project Exclusions (Out of Scope)

Project Goals

- **Nature:** The high-level benefits the project should generate.
- **Scope:** Defines the overarching purpose and organizational alignment.
- **Language:** Broad, visionary, strategic.

Project Objectives

- **Nature:** The specific milestones or steps needed to complete the goals.
- **Scope:** Highly tactical, outlining the exact results to be achieved (scope, schedule, cost, quality).
- **Language:** Simple, digestible, heavily quantified.

Success Criteria correspond to each objective to mathematically or factually prove it has been met.



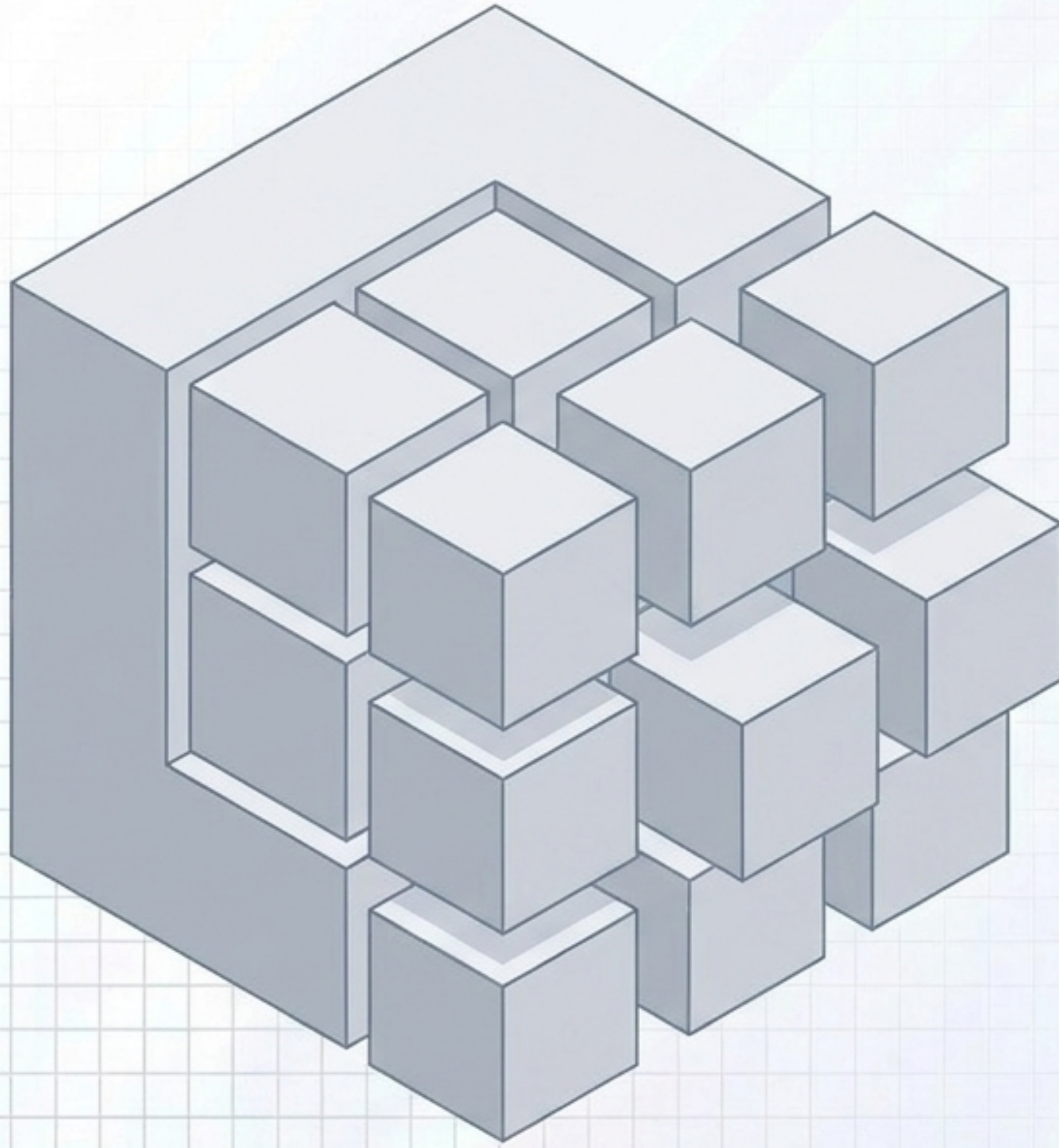
Success Criteria correspond to each objective to mathematically or factually prove it has been met.

The Prompt - Poor Objective

Make the new software faster.

The Solution - SMART Objective





The Anatomy of a WBS

Rule 1: Hierarchical Decomposition

Sub-dividing major deliverables into smaller, manageable components for accurate cost and duration estimation.

Rule 2: Deliverables, Not Actions

A WBS is a deliverable-oriented grouping defining the entire scope. It focuses on the output (e.g., 'UI Design'), not the action (e.g., 'Designing the UI').

Rule 3: The 100% Rule

The WBS must include ALL work defined by the project scope and capture 100% of the deliverables.

Level 1: Project Name

The overarching endeavor (Top-level).

Level 2: Major Deliverables / Phases

The largest chunks of work (e.g., Initiation, Planning, Execution).

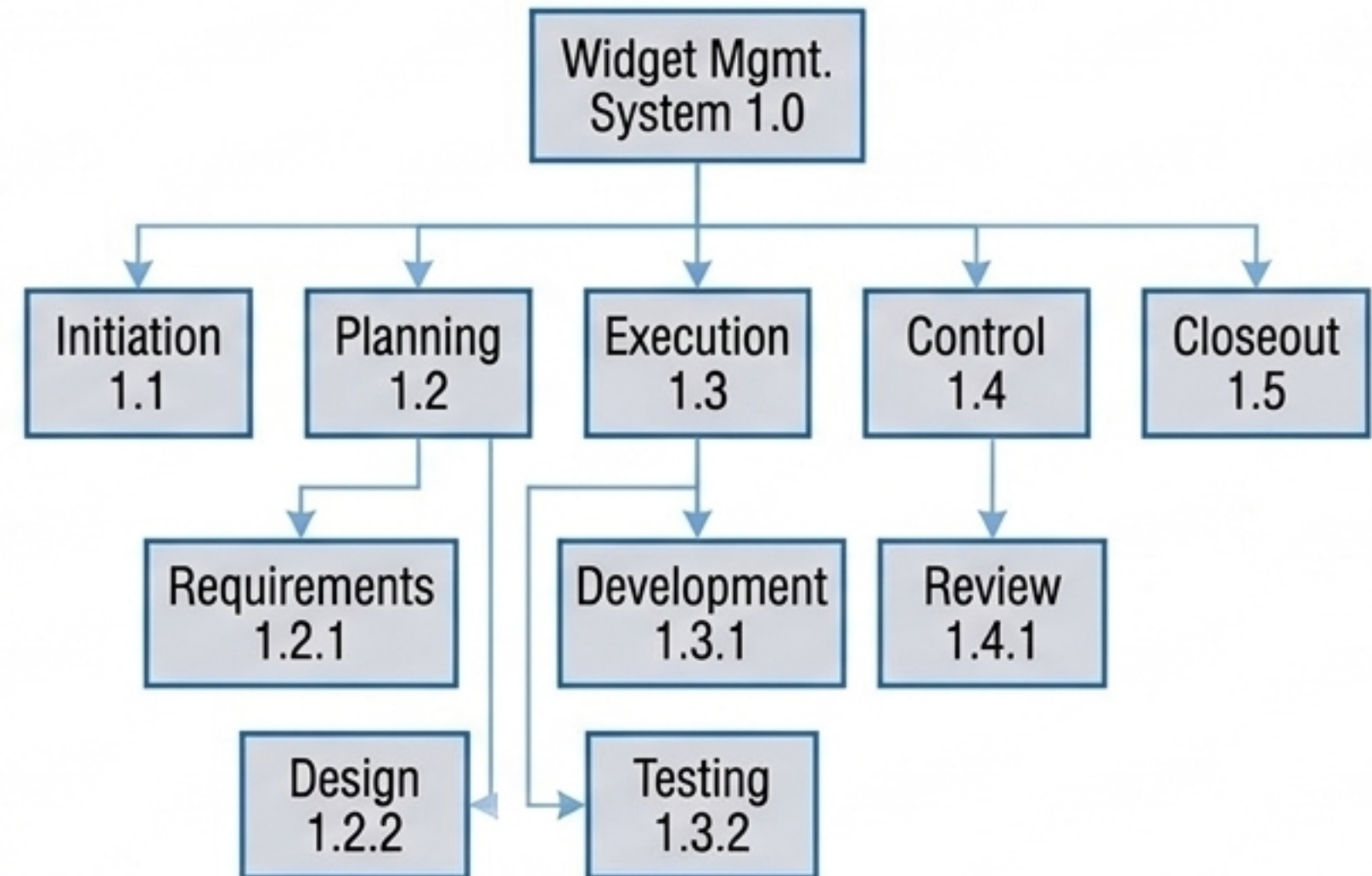
Level 3: Sub-deliverables

The components required to complete a Level 2 phase.

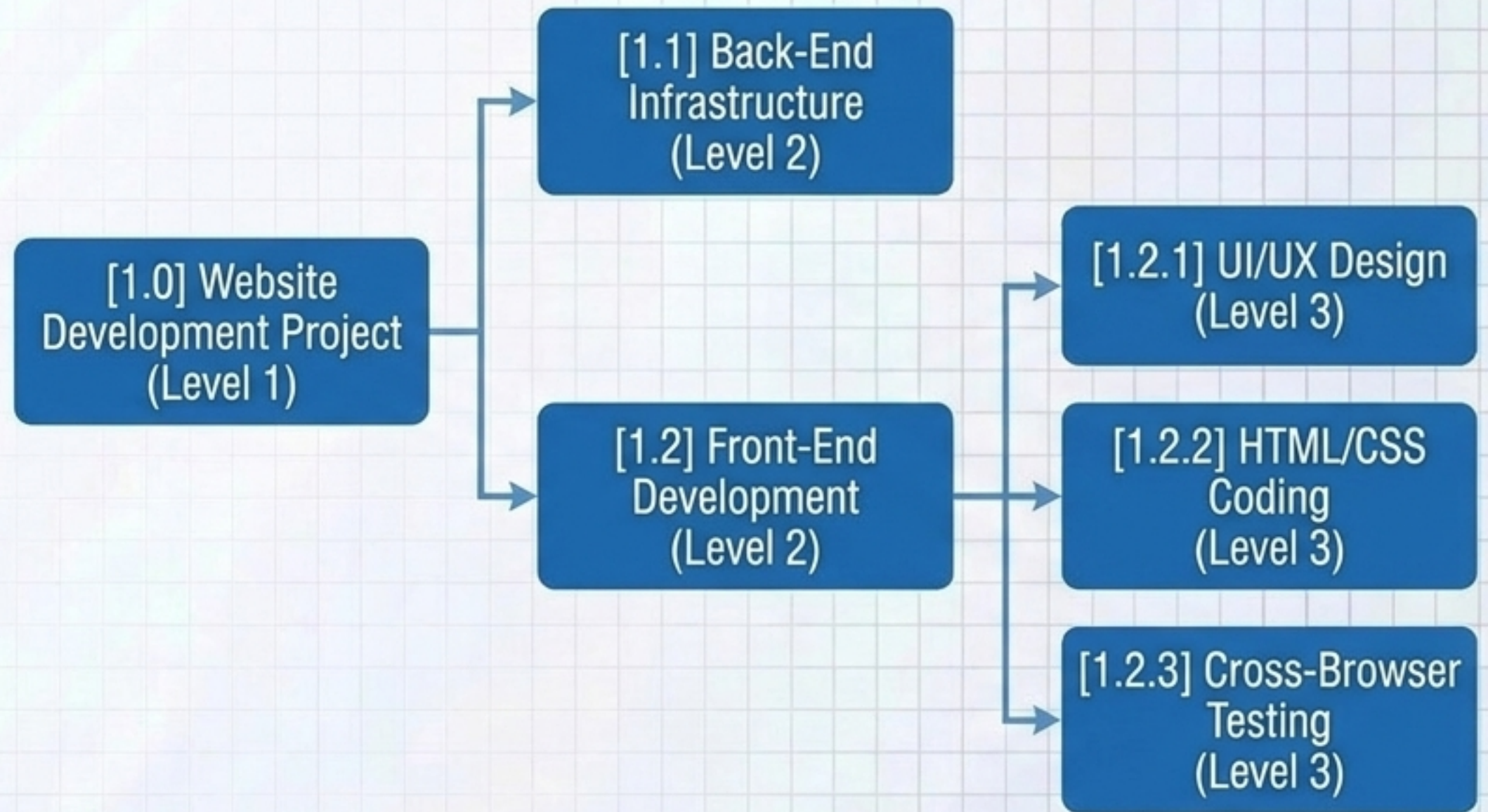
Level 4 & Below: Work Packages

The lowest level, easily assignable to a specific team or individual with clear effort hours.

The Anatomy of a WBS



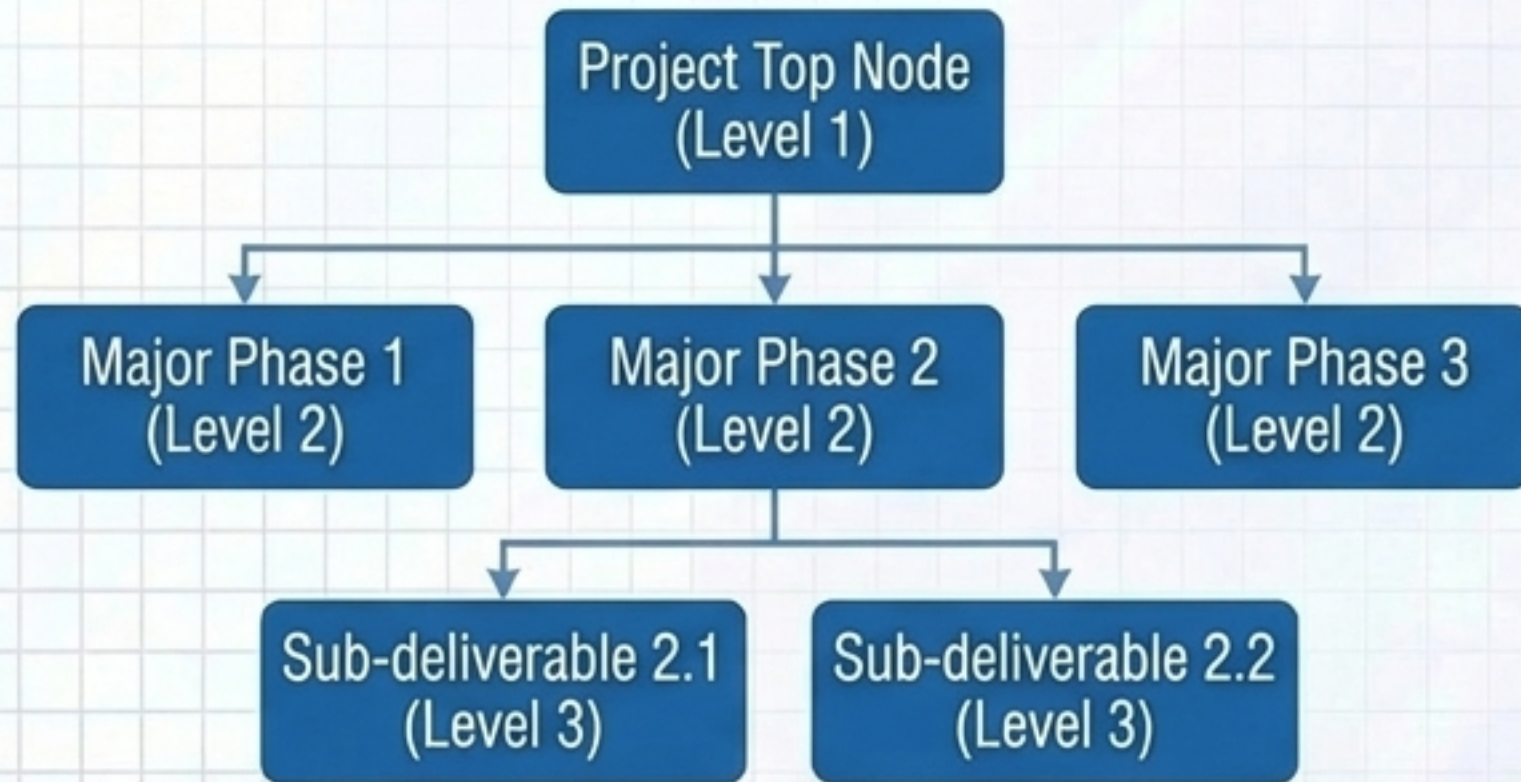
Practice Workbook: Map a 3-Level WBS for a new Website Development Project focusing on Front-End deliverables.



Notice how Level 4 (Work Packages like 'Wireframes') would branch directly off Level 3.

Formats & The WBS Dictionary

Tree Diagram



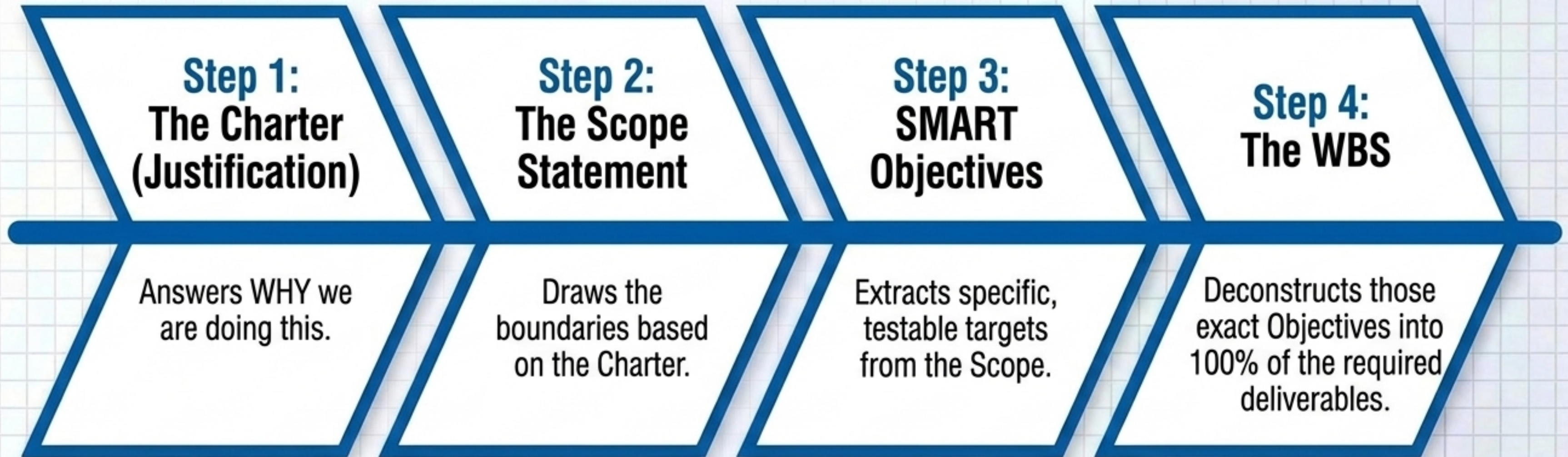
Tree Diagram: Best for visual breakdown and high-level stakeholder communication.

Outline View / Dictionary

TASK NAME	WBS Code	Resources	Dependencies	Effort
Setup Development Environment	3.1	Developer, IT Support, IDE Software	2.3	40 Hours

Outline View / Dictionary: Best for detailed execution. Includes numbering, task names, resource assignments, and effort tracking.

The Golden Thread: Continuous Project Structure



Key Takeaway: A project fails when the thread breaks. A WBS without clear SMART objectives leads to scope creep; objectives without a justified charter lead to misaligned business goals. The structure is the strategy.